Group Activity: Finding NLP Applications by Comparing the Broadway Map and HealthMap

1. Original Source:

- Maps compared: the Broadway map by Dr. John Snow (cholera) and HealthMap.

- Type: Group activity.

- Purpose: Compare the two maps to find NLP applications that can be used to draw conclusions about disease outbreaks.

2. Approach:

- Studied how the Broadway map plotted cholera cases to show where the outbreak was concentrated.

- Studied how HealthMap shows disease outbreak information on a map using reports from online sources.

- Compared the two maps to see where language processing could add value.

- Identified two NLP applications from the comparison.

3. NLP Applications Identified:

Application 1: Rumour Detection Based on Pandemic

- Idea: Detect false or unverified claims about a disease that spread through news and social media during a pandemic.

- How NLP helps: Classify posts and headlines as rumour or reliable information, and track how a rumour spreads.

Application 2: Outbreak Radar Using NLP and Transport Data

- Idea: Combine outbreak information extracted from text with transport data to spot where an outbreak may spread next.

- How NLP helps: Extract disease names and locations from reports, then link them with travel routes to estimate the spread.

4. Skills Required:

For Rumour Detection:

- Text preprocessing: tokenization, stopword removal, stemming and lemmatization.

- Feature extraction: bag of words and TF-IDF.

- Text classification: Naive Bayes and other classifiers.

- Sentiment analysis of posts and headlines.

- Model evaluation: precision, recall and F1 score.

- Python, NLTK and Scikit-learn.

For Outbreak Radar:

- Named Entity Recognition: finding diseases, places and dates in text.

- Information extraction from news and health reports.

- Location mapping: matching place names to map coordinates.

- Working with transport data: routes, travel links and connections.

- Data integration: combining text-based data with transport data.

- Map-based data visualization.

- Python, NLTK, Pandas and Scikit-learn.

Common to both:

- Teamwork, research and critical thinking.

- Comparing sources and drawing conclusions from data.

5. Conclusion:

Comparing the Broadway map and HealthMap showed that placing information on a map reveals patterns in how diseases spread. NLP can extend this idea by detecting rumours during a pandemic and by building an outbreak radar that combines text and transport data.